Hunter-Gatherers and Play

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Analysis of the anthropological literature suggests that hunter-gatherers use play and humor, more or less deliberately, to make their highly egalitarian mode of existence possible. Their methods of governance and sharing, religious beliefs and practices, and productive work are playful; and their children educate themselves through play.

The focus in this article is on *band* hunter-gatherers, also known as *immediate-return* or *egalitarian* hunter-gatherers (as contrasted with the more sedentary, hierarchically organized hunter-gatherer groups referred to variously as *collector*, *delayed-return*, or *non-egalitarian* hunter-gatherers). Wherever they are found, band hunter-gatherers (hereafter referred to simply as hunter-gatherers) live in small, mobile bands that move regularly from place to place within large but relatively circumscribed areas; do not condone violence; are egalitarian in social organization; make decisions by consensus; own little property; readily share what they do own; and have little occupational specialization except those based on gender (Kelly, 1995).

The pure hunter-gatherer way of life is now largely extinguished, pushed out by agriculture, industry, and modern ways generally. But throughout much of the last half of the 20th century, anthropologists could find and study hunter-gatherers who had been very little affected by modern ways. Examples of such groups are the *Ju/'hoansi* (also called the *!Kung*, of Africa's Kalahari Desert), *Hazda* (of Tanzanian rain forest), *Mbuti* (of Congo's Ituri Forest) *Aka* (of rain forests in Central African Republic and Congo), *Efé* (of Congo's Ituri Forest), *Batek* (of Peninsular Malaysia), *Agta* (of Luzon, Philippines), *Nayaka* (of South India), *Aché* (of Eastern Paraguay), *Parakana* (of Brazil's Amazon basin), and *Yiwara* (of the Australian Desert). Although these and other hunter-gatherer groups still exist, the cultures have changed and many of the practices described in this article have been modified or obliterated. The present tense throughout his article refers to the "ethnographic present," that is, to the time at which the observations were made.

Throughout this article the terms "play" and "playful" refer to activity that is (1) self-chosen and self-directed; (2) intrinsically motivated; (3) structured by mental rules; (4) imaginative; and (5) produced in an active, alert, but non-stressed frame of mind. Playfulness can vary in degrees; the more fully an activity contains all of these characteristics, the more playful it is considered to be. For elaboration on this 5-characteristic definition see Definitions of Play.

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Social Play as a Mode of Governance in Hunter-Gatherer Bands

Most hunter-gatherers, wherever they have been studied, live in bands of about 20 to 50 people each, counting children as well as adults. Each band moves as needed to follow the available game and edible plants. At each campsite to which they move, each family builds, from natural materials, a small, temporary hut, the construction of which usually takes just a few hours. Because the band moves frequently, material goods beyond what a person can easily carry are a burden, so there is very little accumulation of property. Each band is an independent entity. The people within the band make all of the band's decisions.

Nearly all researchers who write about hunter-gatherer bands emphasize the extraordinarily high value they place on individual autonomy. Hunter-gatherers' sense of autonomy is different from the individualism of modern Western capitalist cultures. Western individualism tends to pit each person against others in competition for resources and rewards. It includes the right to accumulate property and to use wealth to control the behavior of others. In contrast, as Tim Ingold (1999) has most explicitly emphasized, hunter-gathers' sense of autonomy connects each person to others, in a way that does not create dependencies. Their autonomy does not include the right to accumulate property, to use power or threats to control others, or to make others indebted to oneself. It does, however, allow people to make their own day-to-day and moment-to-moment decisions about their own activities, as long as they do not violate the band's implicit and explicit rules. For example, individual hunter-gatherers are free, on any day, to join a hunting or gathering party or to stay at camp and rest, depending on their own preference.

Intimately tied to hunter-gatherers' sense of autonomy is what Richard Lee (1988) has called their "fierce egalitarianism." Egalitarianism, among hunter-gatherers, goes far beyond the western notion of equal opportunity. It means that nobody has more material goods than anyone else, that everyone's needs are equally important, and that nobody considers himself or herself superior to others. Such equality is part and parcel of hunter-gatherers' autonomy, as inequalities could lead those who have more to dominate those who have less. Hunter-gatherers, of course, recognize that

some people are better hunters or gatherers than others, some are wiser than others, and so on, and they value such abilities. However, they react strongly against any flaunting of abilities or overt expressions of pride.

From an economic point of view, the primary purpose of the band is sharing. The people share their skills and efforts in obtaining food, defending against predators, and caring for children. They also share food and material goods. Such sharing, presumably, is what allowed hunter-gatherers to survive, so long, in challenging conditions. The hunter-gatherer concept of sharing is different from our Western concept. For us, sharing is a praiseworthy act of generosity, for which a "thank you" is due and some form of repayment may be expected in the future. For hunter-gatherers sharing is not a generous act, nor an implicit bargain, but a duty. People are not thanked or praised for sharing, but would be ridiculed and scorned if they failed to share. Anthropologists refer to such sharing as "demand sharing." Failing to share, if you have more than someone else, is a violation of a fundamental rule of hunter-gatherer societies (Ingold, 1999; Wiessner, 1996).

Hunter-gatherers do not have "big men" or chiefs, of the sort common in collector and primitive agricultural societies, who tell people what to do. Some hunter-gatherer groups have no regular leader at all. Others have a nominal leader who speaks for the band in dealing with other bands, but that person has no more formal decision-making power than anyone else. Decisions that affect the whole band, such as that to move from one camp to another, are made by group discussions, which may go on for hours or even days before action is taken. Women as well as men take part in these discussions. Within any given band some people are known to have more wisdom or better judgment than others, and are therefore more influential than others; but any power they exert comes from their ability to persuade and to find compromises that take everyone's desires into account (see, for example, Silberbauer, 1982).

The goal of such discussion is to reach consensus among all who care about the decision. It usually makes no sense to act, as a band, until all band members are ready to go along with the action. Those who are not ready to go along may leave the band, or they may stay as disgruntled members; in either case the band would be weakened. To accept a decision that is strongly rejected by some members is, implicitly, a decision by the band that it would be okay for those members to leave.

Sometimes anthropologists write about hunter-gatherer social life—its individual autonomy, equality, sharing, and consensual decision-making--as if nothing comparable to it exists in Western cultures. The suggestion supported in what follows here, however, is that something quite comparable does exist, in every well-functioning group of people playing a social game.

Comparison of a Hunter-Gatherer Band to a Play Group

Imagine a neighborhood group playing together. To make the example more specific, imagine a game of baseball—not a little-league game run by coaches and umpires, which is not fully play, but a mixed-age pickup game run by the players themselves. The stated goal of each player might be to win, but the real goals are to keep the game going, play well (as defined by each person's own standards), and enjoy a shared activity. The score might be kept, but in the end nobody cares about the score. Even though the game is nominally competitive, it is really a cooperative game in which all of the players, regardless of which team they are on, strive together to make the game last and to keep it fun. Players may even move from one team to another, to keep the teams balanced, as the game progresses. So, it is appropriate to think of all of the players as one play group, not two teams pitted against one another.

A basic characteristic of any social game, if it is really play, is that participation is optional; anyone who wants to leave can do so at any time. Freedom to quit is the most basic freedom in all true play (see Definitions of Play). Since the game requires a certain number of players, everyone who wants the game to continue is motivated to keep the other players happy so they don't leave. This has a number of implications, which are intuitively understood by most players.

One implication is that players must not dominate or bully other players, because people who feel dominated will quit. Another implication is that players must attempt to satisfy the needs and wishes of all the other players, at least sufficiently to keep them from quitting. In this sense, each person, regardless of ability, must be deemed equally worthy. If Marc, Mike, and Mary all want to pitch, the team might let each have a turn at pitching, even though their chance of winning would be better if Henry did all the pitching. Whoever is pitching, that person will almost certainly throw more softly to little Billy, who is a novice, than to big, experienced Jerome. When Jerome is up, the pitcher shows his best stuff, not just because he wants to get Jerome out, but also because anything less would be insulting to Jerome. The golden rule of social play is not, *Do unto others as you would have them do unto you*. Rather, it is, *Do unto others as they would have you do unto them*. The equality of play is not the equality of sameness, but the equality that comes from granting equal validity to the unique needs and wishes of every player.

In any given pick-up game, some people will be better players than others. There will be a tendency for the better players to dominate—to make all the rules, to give orders to others, and so on. However, if they do that, or do it too obviously, the others will quit. So, to the degree that the better players lead, they must learn to do so without dominating, without destroying the other players' sense of choice. The better players must also be careful not to flaunt their superior play. If they flaunt their ability, others may feel belittled and may quit. To keep the game going, players who intuitively understand these rules of play may develop leveling strategies, aimed at preventing anyone from flaunting their ability or behaving in a domineering manner. For example, such displays may be ridiculed, or mocked, with the aim of bringing the overly proud person down a peg or two.

Sharing is also crucial to the game. Some people may come with a baseball glove and/ or a bat, and others may come with nothing. An implicit rule is that all such materials are common property for the duration of the game. The catcher will use whatever catcher's mitt is available, the fielders will use whatever gloves are available, depending in part on the position they are playing, and each batter is free to choose from any of the bats.

As the game progresses, rules may be modified at any time, to make the game more fun and allow it to continue. Anyone can propose a new rule, but to become a rule all players must accept it. In other words, decision-making in social play is by consensus. Consensus doesn't mean that everyone has to agree that the new rule is the best rule possible. It only means that everyone consents to the rule, that is, they are happy enough with it that they aren't going to walk away from the game because of it. Often a great deal of discussion and compromise is required to reach such consensus. A simple majority vote wouldn't suffice, because in that case the minority might feel unhappy and quit; and, again, if too many quit the game is over.

In sum, the key elements that underlie social relationships and governance in a well-operating social game are (1) voluntary participation, with attendant freedom to quit at any time; (2) allowance for much individual autonomy, within the rules of the game; (3) equal treatment of all players, not in the sense of treating them all the same, but in the sense of taking their needs equally into account; (4) obligatory sharing of game-related materials; and (5) consensual decision making. Of these characteristics, the first is the most basic. The freedom of each player to quit is what ensures that those who want the game to continue will behave in ways consistent with the remaining four elements. If players were compelled to stay in the game, then

the more powerful players could dominate, and the autonomy, equality, sharing, and consensual decision-making would be lost.

The five just-listed characteristics of a group playing a game are precisely the elements that anthropologists refer to repeatedly in their discussions of social relationships and governance in hunter-gatherer bands. The meanings of voluntary participation, autonomy, equality, sharing, and consensus within a hunter-gatherer band are quite comparable to their meanings in social play.

Hunter-gatherers are highly mobile not just in the sense of whole bands moving from place to place, but also in the sense of individuals and families moving from band to band. Bands are not permanent structures with fixed memberships. Everyone has friends and relatives in other bands, who would welcome them in. Because of this, and because they are not encumbered by property, individuals may move at a moment's notice from one band to another. People move from band to band for marriage, but they also move to get away from conflicts or simply because they are more attracted to the people or the procedures that exist in another band. Disgruntled groups of people within any band may also, at any time, leave the original band and start a new one. Thus, the decision to belong to any given band is always a person's choice (Ingold, 1999; Woodburn, 1982). The freedom of band members to leave sets the stage for the other playlike qualities of hunter-gatherer life.

Although hunter-gatherers are free at any time to leave, they recognize the value of keeping a band together. The band is the economic and work unit, as well as the social unit, of hunter-gatherer life. A band with stable membership, in which people know one another well and have a history of cooperating, is more valuable than an unstable band. Moreover, people develop close friendships with others in their band. Therefore, members of a band—like participants in a play group—are motivated to behave toward others in ways designed to keep the group together, and this lays the foundation for hunter-gatherers' autonomy, equality, sharing, and consensual decision-making.

Uses of Humor in Hunter-Gatherer Governance

Researchers who have lived in hunter-gather bands often write about the good humor of the people—the joking, good-natured teasing, and laughter. Such humor, which is also common among people everywhere in social play, no doubt serves a bonding function. Laughing together helps create a feeling of closeness and shared identity. Good-natured teasing is a way of acknowledging yet accepting one another's flaws.

Some anthropologists have pointed out that hunter-gatherers' use humor also for another purpose, that of correcting or punishing those who are disrupting the peace or violating a rule. For example, Colin Turnbull (1961) wrote: "[The Mbuti] are good-natured people with an irresistible sense of humor; they are always making jokes about one another, even about themselves, but their humor can be turned into an instrument of punishment when they choose." Similarly, Elizabeth Marshall Thomas (2006) noted that the Ju/'hoansi would not criticize people directly, but would do so through humor. She wrote (p 218): "The criticized person was not supposed to take offense at the jokes and would be sure to laugh along with the others. On the very rare occasions when self-control broke down, such as happened when two women could not stop quarreling, other people made a song about them and sang it when the arguments started. Hearing the song, the two women felt shamed and fell silent. Thus the community prevailed without mentioning the problem directly."

Richard Lee (1988) has commented extensively on hunter-gatherers' use of humor as a tool to quell budding expressions of individual superiority. He wrote (p 261): "There is a kind of rough good humor, putdowns, teasing, and sexual joking that one encounters throughout the foraging world. ...

People in these societies are fiercely egalitarian. They get outraged if somebody tries to put on the dog or to put on airs; they have evolved—independently, it would seem—very effective means for putting a stop to it. These means anthropologists have called 'humility-enforcing' or 'leveling' devices: thus the use of a very rough joking to bring people into line."

As an example, Lee described the practice of "insulting the meat," which occurs among the Ju/'hoansi and various other hunter-gatherer groups. When a hunter brings in a large kill, he must act humbly. If he fails to do so, the others in the band—often led by the grandmothers—will talk about how skinny the animal is or how inept the hunter is, or will in other ways make jokes or songs designed to deflate the hunter's ego. As one of Lee's confidants among the Ju/'hoan explained (Lee, 1988, p 52): "When a young man kills much meat, he comes to think of himself as a big man, and he thinks of the rest of us as his inferiors. We can't accept this. We refuse one who boasts, for someday his pride will make him kill somebody. So we always speak of his meat as worthless. In this way we cool his heart and make him gentle."

The effectiveness of humor as a leveler and reducer of aggression derives from its direct relationship to play. To make fun of something is, in effect, to say, "This thing that you are so proud of, or this dispute that has you so angry, is not as important as you think it is. This is play, and the important thing in play is to be a good sport." When hunter-gatherers use humor to resolve even the most serious social problems they face, they seem to bring all of social life into the domain of play.

The relationship between laughter and play lies deep in our biological makeup. Laughter originated, in primate evolution, as a signal to accompany play fighting. To distinguish play fighting from real fighting, so that a playful attack is not responded to with a real one, players of any species must use some signal to assure one another that their attacks are playful. In monkeys and apes, the play signal is the relaxed open-mouth display, or play face, characterized by a widely open mouth with lower jaw dropped and relatively little tension in the facial muscles. In chimpanzees the play face is often accompanied by a vocalized ahh ahh ahh, which sounds like a throaty human laugh. Such observations leave little doubt that the play face is evolutionarily related to human laughter (van Hoof, 1972). Play fighting and the signals accompanying it constitute the original form of humor. When hunter-gatherers, or humans anywhere, use humor to quell a fight or deflate a puffed-up ego, they are calling on a very primitive mammalian mechanism. They are saying, in effect, "This is play; and in play we don't really hurt anyone and we don't act in a domineering manner." They are saying it in a way that works because it strikes at the gut level of instinct, rather than at the intellectual level of verbal criticism or argument, which people everywhere are good at refuting or ignoring.

And so, by using humor as a means to promote humility and peace, hunter-gatherers capitalize on the instinctive connection between humor and play. Those who are criticized through humor have three choices: They can join the laughter, thereby acknowledging implicitly the foolishness of what they have done, which puts them immediately back into the social game. They can feel and express shame for acting in a way that led to the ridicule, which brings them back into the good graces of the others and allows them more gradually to re-enter the game. Or, they can stew in resentment until they either leave the band or decide to change their ways. A great advantage of humor as a prod behavioral reform is that it leaves the punished persons free to make their own choices and does not automatically end their sense of autonomy and play, as would happen if the punishment involved incarceration, physical violence, or forced banishment.

Rules for Sharing in the Social Game of Life

All social play involves socially shared rules. The overarching purposes of the rules for any social game, if it is truly play, are to coordinate the activities of all of the participants into a coherent whole and to make the game fun for all. The rules require that people resist their natural urges or instincts and exert self-discipline. Much of the joy of social play comes from such exertion and from the aesthetics of taking part in a coordinated, rule-restrained social activity. All this, which can be said about the rules of every form of social play, can also be said about the rules governing any hunter-gatherer society. For example, hunter-gatherers everywhere have rules for distributing foods and sharing the few material goods they own. The goal, always, is material equality, which may be essential for the band's survival, but the means of achieving that goal are often quite elaborate and playlike. Consider, as illustration, the rules for distributing meat.

When hunters bring a large kill into the camp, it is a time of general rejoicing. The only person who cannot rejoice is the hunter who killed the animal; he must behave modestly and act as if the animal is skinny and worthless. The meat from the kill is then distributed to families and individuals in the camp in a manner that follows a game-like set of rules, though the rules differ from society to society. One rule specifies who may carve up the meat and distribute it in the first wave of distribution. Among the Ju/'hoansi, the official initial "owner" of the meat, who has the right to distribute it, is not the hunter but the person who owned the arrow that killed the animal. There is much giving and lending of arrows, among all members of the band, women as well as men, so anyone might own an arrow and lend it to a successful hunter (Thomas, 2006). Some other hunter-gatherer societies likewise attribute initial game ownership to the person who owned the implement (such as arrow, poisoned dart, or net) that was used to make the kill or capture (Wiessner, 1996). Such rules assure that even the good will that is generated by the distribution of meat does not go just to successful hunters, but is distributed throughout the band.

In apparently all hunter-gatherer groups there is no economic advantage in being the distributor of meat. That person is never allowed to take a larger share than anyone else, and often he must take a smaller share. Some societies have explicit rules for the order of distribution. Among the Yiwara, for example, the man who brings home a kill must give the first and best portions to those who are least closely related to him by blood, including his in-laws, and must leave for his immediate family and himself the least desired portion (Gould, 1969). Among the Hazda, pregnant women are given first priority (Weissner, 1996). All these rules seem to have practical purposes, but the ceremonial spirit in which they are followed seems to put them at least partly into the realm of play.

In relatively large hunter-gatherer bands, the distribution of meat occurs in waves. The first wave involves distribution among a pre-designated set of adults, who then distribute those portions among others, who in turn distribute the portions they received. The end result is that everyone receives roughly equal portions, with some differences depending on perceived need. Kirk Endicott (1988) points out that food sharing among the Batek may continue even when everyone has plenty. Families may give portions of food to others who already have adequate portions and may receive, from others, the same kinds of foods that they have just given away. Here the implicit rules of sharing clearly go beyond the practical purpose of making sure that everyone gets a fair share. The means (sharing even when people already have equivalent portions) here take precedence over the end, which makes the sharing playlike.

So crucial are the rules of food sharing to hunter-gatherer bands that anyone who fails to share is, in essence, opting out of the game, declaring that he or she is no longer a member of the band. Kim Hill (2002), concerning the Aché, wrote, "... it is my impression that those who refuse to share game would probably be expelled from the band." The analogue to this, in a pick-up game of baseball, would be the kid who, when he gets the ball, just holds

on to it and refuses to throw it to anyone else.

Even more game-like is the sharing of materials other than food. Hunter-gatherers own little, and the objects they do own, such as beaded decorations and tools, have limited value because they are made from readily available materials and can be replaced without great trouble by band members who are highly skilled at making them. Yet, the people cherish such objects, not as treasures to hoard but as potential gifts to others. Such objects are circulated in continuous rounds of gift giving, which promote friendships. People in collector and agricultural societies also often have gift-giving traditions, but in those societies the giving may take on competitive, power-assertive, and dependence-producing functions (Hayden, 1996). In contrast, hunter-gatherers take pains to keep their gift-giving modest, friendly, non-competitive, and in those senses playlike.

The Ju/'hoansi, for example, have a formal gift-giving system, referred to as hxaro. Each Ju/'hoan adult has roughly 10 to 20 regular hxaro partners, most of whom live in other bands, sometimes more than 100 miles away. Each person travels regularly, by foot, to visit his or her hxaro partners and present them with gifts. Giving between any pair of partners always goes in both directions, but care is taken to prevent the giving from looking like trade. Gifts are never reciprocated immediately, and there is no expectation that the gifts balance out to be equal in value. Each gift is given and received in a spirit of friendship, not as something owed to the other. Hxaro partners are said by the Ju/'hoansi to "hold each other in their hearts" (Weisner, 2002).

By having hxaro partners in many different bands, spread out over large areas, the Ju/'hoansi protect themselves from complete dependence on their own band and location. They are welcomed, for as long as they wish to stay, wherever they have such a partner. So, what at first seems to be wasted effort—walking hundreds of miles a year to deliver gifts that have little material value—is actually a socially valuable game. It helps maintain peace between bands, and it frees people from the confinement and possibility of exploitation that would result if they could not move freely from one band to another. It also facilitates marriages between people of different bands, which is essential among all hunter-gatherers to prevent inbreeding. But these ultimate gains are not the immediate, conscious motives for most of the visits. The conscious motives are to experience the joys that come from visiting old friends, presenting them with gifts, and following the rules of a life-long game.

A Playful Approach to Religion

A case can be made that religious faith, everywhere, taps into the human capacity for play. Faith is belief that does not require empirical evidence. To believe without evidence is to make-believe. In any social game the players accept, for the purpose of the game, the fictional premises that provide the game's context. Jill is the princess, Johnny is the fierce dragon, and the couch is a bridge with a troll living under it. Only during time out can Jill and the others say that they were merely pretending. It can be argued that religion, for the devout, is play for which there is no time out. (A similar point was made by Huizinga, 1955.)

If we think of social life as a grand human game, then the religious beliefs of a society provide a context for understanding the goals and rules of the game and for making decisions. The religious beliefs both reflect and help to support the society's socioeconomic structure. From this point of view it is no surprise that monotheistic religions that blossomed in feudal times portray a hierarchical view of the cosmos, with an all powerful God, "king of kings," at the top, and a storyline focused on requirements of obedience and service to lords and masters. It is also no surprise that hunter-gatherer

religions reflect an egalitarian view of the spirit world, populated by a multitude of deities, none of whom has authority over the others or over human beings.

Because of their egalitarian foundation, hunter-gatherer religions are playful in ways that go beyond the general way in which all religions can be thought of as play. For devout Jews, Christians, and Muslims, the cosmos is imbued with serious moral purpose, to which humans must bend in ways that run counter to the spirit of play. For hunter-gatherers, in contrast, the cosmos is capricious. The hunter-gatherer deities themselves are playful and even comical beings, not stern judges. Their interactions with people can most often be described as whimsical. A deity may hurt or help a person just because he or she feels like it, not because the person deserves it, and in that sense, at least, the deities are personifications of natural phenomena such as the weather. A common character in the hunter-gatherer spirit world is what mythologists call the "trickster" (Guenther, 1999). The trickster is typically a partly clever, partly bumbling, morally ambivalent being who manages to interfere with the best-laid plans of the other deities and humans. The trickster character is not necessarily represented in just one deity; it may be an aspect of personality that runs through most or all of them.

One of the Ju/'hoan deities has characteristics that might, at first, lead us to view him as equivalent to the single god of modern monotheistic religions. This deity, called Gao Na, is the creator of the universe. First he created himself and the other deities; then the earth, water holes in the earth, and water to fill the holes; then the sky, sun, moon, stars, rain, wind, lightning, plants, animals, and human beings. Yet, despite such power of creation, Gao Na is not seen as particularly powerful in other respects and certainly not as wise. In fact, consistent with their general practice of leveling those who might think too highly of themselves, the Ju/'hoansi delight in portraying Gao Na as a fool (Thomas, 2006). For example, in some of their stories, Gao Na's wives trick him repeatedly into jumping into a pit full of feces. They tell him that there is a fat eland under a pile of branches, and he leaps happily into the pile to get it, only to fall into the pit. Later they tell him another story, about some other prize under the branches, and he jumps in again. Similar stories, apparently aimed at leveling the deities, can be found in other hunter-gatherer religions (e.g. Endicott, 1979).

The religious practices of most hunter-gatherers include shamanic ceremonies. The primary serious purpose of such ceremonies is healing, but the ceremonies also provide an opportunity for people to interact personally, in all sorts of ways, with members of the spirit world. Shamans enter into trance states in which they take on the properties of, and/or communicate with, specific deities. In a review of shamanic activities, Mathias Guenther (1999, 427-428) wrote: "Often the shaman is a showman who employs rich poetic imagery and histrionics. He may sing and dance, trembling and shrieking, and speak in strange languages. He may also employ prestidigitation and ventriloquism. . . . Shamanic séances are very much performance events, not infrequently with audience feedback. They involve the shaman in role playing, engaging in dialogue with various spirits, each of whose counter-roles he plays himself." Among some hunter-gatherer groups the whole band is involved in the dancing, singing, and drumming; all of them, effectively, are shamans or at least contributors to the shamanic experience. When spirits are called forth in such exercises, in apparently any hunter-gatherer group, they are not treated reverently; they are treated much as the people treat one another. The communication may involve mutual joking, teasing, laughing, singing, and dancing, as well as requests for healing.

Anthropologists refer to the shamanic and other religious ceremonies as "rituals," probably because that term has come to be used for any religious ceremony that has some sort of regular structure to it. But the ceremonies are clearly not rituals in the sense of strict, uncreative adherence to a prescribed form. In fact, some hunter-gatherer researchers have claimed that the "rituals" that they observed in the groups they studied were indistinguishable from play (Tsuru, 1998).

A number of researchers have commented that hunter-gatherers, in general, are highly practical people, not much given to magic or superstition (e.g. Bird-David,1992; Thomas, 2006). Shamanic healing might be seen as an exception, but such healing may actually work to the degree that diseases have psychological components. In general, hunter-gatherer religious ceremonies have more to do with embracing reality than with attempting to alter it. As an example, Thomas (2006) describes how the desert-dwelling /Gwi people use their rain dance not to bring on rain but to welcome it joyfully and partake in its power when they see it coming. Gould (1969, p 128), writing of the Yiwara, makes the same point in stating that these people "...do not seek to control the environment in either their daily or their sacred lives. Rituals of the sacred life may be seen as the efforts of man to combine with his environment, to become 'at one' with it." Such ceremonies can be viewed as a form of play in which aspects of the natural world, personified in the deities, become playmates.

On the dimensions that commonly distinguish religious liberals from religious fundamentalists in the West, hunter-gatherers appear everywhere to be at the liberal end. Although hunter-gatherers find meaning in their stories about the spirit world, they do not treat them as dogma. Neighboring bands may tell similar stories in different ways, or may tell different stories, which contradict one another, but nobody takes offense. Hunter-gatherer parents do not become upset when their children marry into another group and adopt religious beliefs and practices that differ from those they grew up with (e.g. Endicott, 1979; Guenther, 1999). To leave one band and join another, with different religious practices, is in this sense like leaving a group who are playing one game and joining another who are playing a different game. There seems to be an implicit acknowledgment that religious stories, while in some ways special and even sacred, are in the end just stories.

Turnbull (1961) contrasts the light-heartedness of Mbuti hunter-gatherers' religious beliefs and practices with the fearful superstitions of the nearby agricultural people. The agriculturalists truly fear the forest spirits, so much so that they rarely venture into the forest. In contrast, while the Mbuti claim to believe in the same spirits and to interact with them in their religious ceremonies, they do not, in their everyday lives, manifest any fear of them. One of their ceremonies involves the playing of the molimo—an enormously long trumpet, traditionally made by hollowing out a log from a molimo tree. The men of a band are keepers of this instrument, and, on special occasions, they bring it out at night. The sound of the molimo is deemed sacred, and women are supposed to be frightened of it and to believe that it comes from a terrible animal spirit. According to Turnbull, when he observed the ceremony, the women played their parts well, staying in their huts and acting frightened. But they were not really frightened; they seemed to know perfectly well that this was all a grand game instigated by the men. Other anthropologists have likewise contrasted the playful attitudes of hunter-gatherers toward their deities with the fearful attitudes of neighboring sedentary people (e.g. Endicott, 1979; Tsuru, 1988).

A Playful Approach to Productive Work

Our word work has two different meanings. It can mean toil, which is unpleasant activity; or it can mean any activity that accomplishes something useful, whether or not it is pleasant. We use the same word for both of these meanings, because from our cultural perspective the two meanings often overlap. By all accounts, hunter-gatherers do not have a concept of work as toil (e.g. Gowdy, 1999). They do not confound productiveness with unpleasantness. They do, of course, engage in many productive activities, which are necessary to sustain their lives. They hunt, gather, build huts, make

tools, cook, share information, and so on. But they do not regard any of this as burdensome. They do these things because they want to. Work for them is play.

How do they manage this? What is it about hunter-gatherer work that makes it enjoyable rather than burdensome? Researchers' descriptions suggest that the following four factors contribute to hunter-gatherers' abilities to maintain a playful attitude toward even those activities that they must engage in to survive:

1. The work is not burdensome because there is not too much of it.

According to several quantitative studies, hunter-gatherers typically devote about 20 hours per week to hunting or food gathering and another 10 to 20 hours to chores at the campsite, such as food processing and making or mending tools (e.g. Lee, 2003; Sahlins, 1972). All in all, the research suggests, hunter-gatherer adults spend an average of 30 to 40 hours per week on all subsistence-related activities combined, which is considerably less than the workweek of the typical modern American, if the American's 40 or more hours of out-of-home work is added to the many hours spent on domestic chores.

One anthropologist, Marshall Sahlins (1972), has famously characterized hunter-gatherer societies collectively as "the original affluent society." An affluent society, by Sahlins's definition, is one in which "people's material wants are easily satisfied." Hunter-gatherers are affluent not because they have so much, but because they want so little. They can provide for those wants with relatively little work, and, as a result, they have lots of free time, which they spend, according to one observer of the Ju/'hoansi (Shostak, 1981, p 10), at such activities as "singing and composing songs, playing musical instruments, sewing intricate bead designs, telling stories, playing games, visiting, or just lying around and resting." These are just the kinds of activities that we would expect of happy, relaxed people anywhere.

2. The work is varied and requires much intelligence, knowledge, and skill.

Play requires mental challenge and an alert, active mind engaged in meeting that challenge (see Definitions of Play). The least play-like work is that which is mind-numbingly repetitive and dull. Hunter-gatherer work is almost always challenging.

Hunting, as it is done by hunter-gatherers, requires great intelligence, knowledge, and physical skill. The hunters have a vast knowledge of the habits of the perhaps two to three hundred different mammals and birds they hunt. They can identify each by its sounds and tracks as well as sight. Louis Liebenberg (1990) has developed the thesis that hunter-gatherers' tracking of game marked the origin of scientific reasoning. Hunters use the marks they see in the sand, mud, or foliage as clues, which they combine with their accumulated knowledge from past experience, to develop and test hypotheses about such matters as the size, sex, physical condition, speed of movement, and time of passage of the animal they are tracking. Moreover, The tools of hunting—which, depending on the culture, might be bows and arrows, blow pipes and poisoned darts, spears and spear throwers, snares, or nets—must be crafted to perfection, with great skill. And great skill is needed, too, in the use of the tools (e.g. Kaplan et al., 2000).

The gathering of vegetable foodstuffs, which is done mostly by women, likewise requires great knowledge and skill. Gatherers must know which of the countless varieties of roots, tubers, nuts, seeds, fruits, and greens in their area are edible and nutritious, when and where to find them, how to dig them (in the case of roots and tubers), how to extract the edible portions efficiently (in the case of grains, nuts, and certain plant fibers), and in some cases

how to process them to make them edible or more nutritious than they otherwise would be. These abilities include physical skills, honed by years of practice, as well as the capacity to remember, use, add to, and modify an enormous store of culturally shared verbal knowledge (Bock, 2005; Kaplan et al., 2000).

3. Most work is done in a social context.

We are highly social beings. We like to be with others of our kind, especially with those we know well; and we like to do what our friends and colleagues do. Hunter-gatherers live very social lives. Nearly all of their activity is public. Most of their work is done cooperatively, and even that which is done individually is done in social settings. Men usually hunt in ways that involve teamwork; and women usually forage in groups. Concerning the latter, Wannenburgh (1979, p 41) wrote, of the Ju/'hoansi bands he studied, "In our experience all of the gathering expeditions were jolly events. With the [Ju/'hoansi's] gift of converting chores into social occasions, they often had something of the atmosphere of a picnic outing with children." A social setting—with cooperative efforts, mutual encouragement, and joking and laughter—always helps promote a playful attitude toward work.

4. Each person is free to choose when, how, and whether to work.

The most crucial ingredient of play is the sense of free choice. Players must feel free to play or not play and must invent or freely accept the rules. In Western cultures, workers who must follow, step-by-step, the directions of a micromanaging boss are the least likely to consider their work to be play. Hunter-gatherers have developed, to what in our culture may seem to be a radical extreme, an ethic of personal autonomy. They deliberately avoid telling each other how to behave, in work as in any other context (e.g. Thomas, 2006). Each person is his or her own boss.

On any given day at a hunter-gatherer camp, a hunting or gathering party may form. The party is made up only of those who want to hunt or gather that day. That group decides collectively where they will go and how they will approach their task. Anyone unhappy about the decision is free to form another party, or to hunt or gather alone, or to stay at camp all day, or to do anything at all that is not disruptive to others. There is no retribution for backing out. A person who doesn't hunt or gather will still receive his or her share of whatever food is brought back. By adopting this strategy, hunter-gatherers avoid being held back, in their foraging, by someone who is there only begrudgingly and has a bad attitude about it. And because they adopt this strategy, all members of the band can experience their hunting and gathering as play. Such freedom does open up the possibility of free riding by individuals who choose not to hunt or gather over an extended period, but such long-term shirking apparently happens rarely if at all (Endicott, 1988; Hawkes, 1993). It is exciting to go out hunting or gathering with the others, and it would be boring to stay in camp day after day.

Play as the Route to Education

Education is essential to the human condition. People everywhere depend, for their survival, on skills, knowledge, and ideas that are passed from generation to generation; and such passing along is, by definition, education. This is as true of hunter-gatherer cultures as it is of our own. Hunter-gatherer adults, however, don't concern themselves much with the education of children. They assume that children will educate themselves through their own, self-directed exploration and play.

Our Western cultural notions of education, and of childcare in general, are founded on agricultural metaphors. We speak of raising children, much as

we speak of raising chickens or tomatoes. We speak of training children, much as we speak of training horses. Our manner of talking and thinking about parenting suggests that we own our children, much as we own our domesticated plants and livestock, and that we control how they grow and behave. Just as we train horses to do the tasks that we want them to do, we train children to do the tasks that we think will be necessary for their future success. We do that whether or not the horse or child wants such training. Training requires suppression of the trainee's will, and hence of play.

Hunter-gatherers, of course, do not have agriculture, and so they do not have agricultural metaphors. In their world all the plants and animals are wild and free. Young plants and animals grow on their own, guided by internal forces, making their own decisions. And that, briefly, is the attitude that hunter-gatherer adults take toward children's education. The adults provide the environment that makes children's education possible, but they do not experience any need to direct or motivate it. Hunter-gatherers' treatment of children is very much in line with their treatment of adults. They believe that each person's needs are equally important and that each person, regardless of age, knows best what his or her own needs are.

Anthropologists often describe hunter-gathers' style of childcare as indulgent, though a better term might be trusting. The adults trust and therefore indulge children's instincts, including their instincts to play and explore. They believe that children know best what they need and when they need it, so there are no or few battles of will between adults and children (e.g. Barry et al., 1959; Konner, 2005). As one group of researchers (Gosso et al., 2005) wrote, "[Hunter-gatherers adults] do not interfere with their children's lives. They do not beat, scold, or behave aggressively with them, physically or verbally, nor do they offer praise or keep track of their development."

In Western cultures parents often complain about grandparents and other kin who undermine parental discipline and "spoil" the child. Among hunter-gatherers, such parental discipline is apparently not possible, even if it were desired, because other adults in the band would always undermine it. The result of such practices is that hunter-gatherer children are self-assertive and self-controlled, but not "spoiled," at least not spoiled from the perspective of hunter-gatherer values. A propos of this, Thomas (2006, p 198-199) wrote: "We are sometimes told that children who are treated so kindly become spoiled, but this is because those who hold that opinion have no idea how successful such measures can be. Free from frustration or anxiety, sunny and cooperative,... the Ju/'hoan children were every parent's dream. No culture can ever have raised better, more intelligent, more likable, more confident children."

Given the indulgence that hunter-gatherer adults exhibit toward children, it is no surprise that the children spend most of their time playing. Play, almost by definition, is what children want to do. The adults have no qualms about this, because they believe that it is through play that children learn what they must to become effective adults. In a survey of ten hunter-gatherer researchers, who had lived in 7 different hunter-gatherer cultures, all of the researchers said that children were free to play essentially from dawn to dusk every day (see Gray, 2009). In a published report on how Ju/'hoan children spent their time, Patricia Draper (1976, pp 205-206) concluded: "[Ju/'hoan] children are late in being held responsible for subsistence tasks. Girls are around 14 years old before they begin regular food gathering and water- and wood-collecting. This is in spite of the fact that they may be married before this age. Boys are 16 years old or over before they begin serious hunting. ... Children do amazingly little work." In a study of peoples with mixed hunter-gatherer and agricultural subsistence, in Botswana, John Bock and Sarah Johnson (2004) found that the more a family was involved in hunting and gathering, and the less they were involved in agriculture, the more time children had to play.

Incorporation of Adult Activities Into Play

Hunter-gatherer children are never isolated from adult activities. They observe directly all that occurs in camp––the preparations to move; the building of huts; the making and mending of tools and other artifacts; the food preparation and cooking; the nursing and care of infants; the precautions taken against predators and diseases; the gossip, discussions, arguments, and politics; the songs, dances, festivities, and stories. They sometimes accompany adults on food gathering trips, and by age 10 or so boys sometimes accompany men on hunting trips. In the course of their daily lives, they see, hear, and have the opportunity to explore everything that is relevant to becoming a successful adult in their culture, and they incorporate all of this into their play. They play at the activities that they observe in the adults around them, and as they grow older, their play turns gradually into the real thing. There is no sharp division between playful participation and real participation in the valued activities of the band.

The above-mentioned survey of researchers elicited many examples of valued adult activities that were mimicked regularly by children in play. Digging up tubers, fishing, smoking porcupines out of holes, cooking, caring for infants, climbing trees, building vine ladders, building huts, using knives and other tools, making tools, carrying heavy loads, building rafts, making fires, defending against attacks from predators, imitating animals (a means of identifying animals and learning their habits), making music, dancing, storytelling, and arguing were all mentioned by one or more respondents. The specific lists varied from culture to culture, in accordance with differences in the skills that were exemplified by adults in each culture. All of the respondents said that boys in the culture they studied engaged in a great deal of playful hunting. The two respondents who studied the Agta—a culture in which women as well as men regularly hunt—noted that girls as well as boys, in that culture, engaged in much playful hunting.

Apparently, when children are free to do what they want, they spend much of their time playing at the very activities that they see, from direct experience, are most crucial for success in their culture. Their conscious motive is fun, not education. It is exciting for children, everywhere, to pretend that they are powerful, competent adults, doing beautifully and skillfully what they see the adults around them doing. From an evolutionary perspective, it is no coincidence that children are constructed in such a way.

Equally important to learning how to hunt and gather, for hunter-gather children, is learning how to interact with others assertively yet peacefully. In their play, children practice arguing. Turnbull has described how older Mbuti children (age 9 and up) playfully rehash and try to improve upon the arguments that they have heard among adults. Here are Turnbull's (1982) words:

"It may start through imitation of a real dispute the children witnessed in the main camp, perhaps the night before. They all take roles and imitate the adults. It is almost a form of judgment for if the adults talked their way out of the dispute the children having performed their imitation once, are likely to drop it. If the children detect any room for improvement, however, they will explore that, and if the adult argument was inept and everyone went to sleep that night in a bad temper, then the children try and show that they can do better, and if they cannot, then they revert to ridicule which they play out until they are all rolling on the ground in near hysterics. That happens to be the way many of the most potentially violent and dangerous disputes are settled in adult life."

The Age-Mixed, Noncompetitive Nature of Children's Play

The play of hunter-gatherer children is not only informed by what they have learned by observing others of all ages in the band, but it occurs almost always in age-mixed groups. Because hunter-gatherer bands are small and births are widely spaced, the number of potential playmates for any given child is limited. Even if hunter-gatherer children wanted to segregate by age, they would rarely find more than one or two playmates within a year or

two of their own age and often none. A typical play group might consist of half a dozen children ranging in age from 4 to 11, or from 9 to 15. As Patricia Draper put it, in her response to the survey: “Any [Ju/’hoan] child with enough motor and cognitive maturity could enter into any game. Older teenagers and adults could and did play as well, though not for as long or with the same enthusiasm as the children.”

Research on age-mixed play in our culture suggests that such play differs qualitatively from same-age play (Gray, 2011; Gray & Feldman, 2004). It is less competitive and more nurturing. When playmates differ greatly in age, size, and strength, there is little point in trying to prove oneself better than another. In such play, older children typically help younger children along, which allows the younger ones to play in more sophisticated ways than they would alone and gives the older ones valuable experience in helping and nurturing.

In the 1950s and ‘60’s, using data from the Human Relations Area Files, John Roberts and his colleagues compared the types of competitive games commonly played in different cultures. One of their conclusions was that the only cultures that seemed to have no competitive games at all were hunter-gatherer cultures (Sutton-Smith & Roberts, 1970). In response to a question about competitive play in the above-mentioned survey, only two of the ten respondents said that they had seen any competitive play in the culture they had studied, and those two said that they had “seldom” seen it.

In the most extensive descriptive account of the play and games of any hunter-gatherer group, Lorna Marshall (1976) pointed out that most Ju/’hoan play is informal and non-competitive, and that even their more formal games, which have explicit rules and could be played competitively, are played non-competitively. For instance, Ju/’hoan children of ages 5 to 15, of both sexes, often play a game of throwing the zeni. The zeni consists of a leather thong, about 7 inches long, with a small weight fastened at one end and a feather at the other. The player hurls it into the air as high as possible with a stick, then tries to catch it with the stick when it comes fluttering down, and from that position hurls it again. The game is played with much skill, and it could be played competitively––for instance, by seeing who can hurl it the highest or catch it the most times in succession––but, according to Marshall, it is not played that way. Players try to do their best, but comparisons to others’ performances are not made.

Another Ju/’hoan game with rules is the melon game, played by women and girls. This game involves singing, dancing, and clapping, all according to specific rules, while simultaneously keeping a small melon moving from one dancer to another by tossing it backward, over one's head, to the next person in line. The goal is to keep everyone in harmony with everyone else and keep the melon moving without dropping it. The game could be played in a competitive manner by saying that anyone who drops the melon is “out,” but it is not played that way. The goal always is cooperation, not competition.

The point of hunter-gatherer play is not to establish winners and losers, but to have fun. In the process of having fun, the players develop skills requiring strength, coordination, endurance, cooperation, and wit, and they solidify their bonds of friendship. If the focus were on competition, the pressure to win could reduce the playfulness and fun of the activity. Instead of cementing friendships, competitive games could produce arrogance in winners and envy or anger in losers, which would weaken rather than strengthen the community.

Concluding Thoughts

One way to think about hunter-gatherers’ uses of play is to suppose that our species, by nature, has two fundamentally opposing ways of structuring

social interactions, which we inherited from our mammalian ancestors. One way is the method of dominance. The literature on mammalian social behavior, particularly that on primate social behavior, is replete with discussions of dominance hierarchies and struggles for status. Dominance hierarchies give structure to the social interactions within animal colonies and prevent the chaos that would occur if each new opportunity for food, or for mating, resulted in a renewed struggle. The other way of structuring social interactions is the method of social play.

Play in the animal world always involves the temporary renunciation of dominance. Most mammalian social play takes the form of playful fighting and chasing. Such actions can remain playful only as long as nobody is hurt and the needs of all participants are met. When two young monkeys or chimpanzees engage in a play fight, the stronger one deliberately self-handicaps, and the "fight" is not a fight in the sense of establishing a winner or loser. The playful "combatants" alternate in taking defensive and offensive positions, and they refrain from using their teeth or other weapons in a manner that could hurt the other (Biben, 1998; Symons, 1978). In playful chases, the two take turns in chasing and being chased, like children playing tag. In all social play, each animal must continuously behave in such a way as to meet the needs of the other, while still satisfying it's own needs. Failure to do that would terminate the game. So, during play, a new sort of relationship emerges between individuals, one that is based not on power assertion, but on power restraint and sensitivity to the needs of the other player.

The main thesis of this article is that hunter-gatherers (of the band variety) everywhere developed cultural practices that combatted the tendency toward dominance by maximizing the tendency to play. Hunter-gatherers' existence apparently requires an intense kind of long-term sharing, which was not based just on blood relationships or direct reciprocity. Such sharing would be destroyed by strivings for dominance. Dominance induces fear and anger, while play induces unity and friendship. The kind of sharing upon which hunter-gatherers depends apparently requires the feelings of unity and friendship that play can produce. Therefore, to survive, hunter-gatherers maximized their playful tendencies. This "play theory" of hunter-gatherer equality can be understood as a complement or supplement to the "reverse dominance" theory developed by Christopher Boehm (1999), which contends that hunter-gathers' equality emerges from the ability of the band as a whole to dominate any individuals who strive to dominate.

In addition to the cultural accentuation of play, it is possible that further biological adaptations increased hunter-gatherers' capacity to develop playful solutions to life's problems. If we assume that the needs for intense sharing were present for hundreds of thousands of years in our human ancestors, then natural selection might have expanded and elaborated upon the play instincts inherited from our earlier primate ancestors. In most mammals, including most primates, play occurs mostly among the young and apparently serves primarily as a means to practice life-enhancing skills. In some primates, however, play may also serve a bonding function, helping to counteract the fear induced by dominance systems and thereby helping to promote cooperation. This may help explain why, in some primates, social play is observed to some degree among adults as well as juveniles (Pellis & Iwaniuk, 2000). A great increase in the need for cooperation and sharing based on friendships could have led to further expansion of the human play drive into adulthood and to an increased flexibility of that drive, allowing it to be applied in a wider variety of contexts. From this point of view, Johan Huizinga's (1955) term Homo ludens (playful man) may well be an apt descriptor for our species.

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